MedGemma-Powered Medical Assistant: A Lightweight Multimodal Clinical Support System with Retrieval-Augmented Generation

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***Abstract*—Recent advances in multimodal foundation models have enabled new possibilities for intelligent healthcare assistance systems. However, many existing medical AI platforms rely heavily on cloud-based infrastructure, limiting accessibility, privacy, and deployment flexibility. This paper presents a lightweight multimodal medical assistant powered by MedGemma-4B and Retrieval-Augmented Generation (RAG) for clinical support applications. The proposed system integrates medical image analysis, clinical question answering, and patient data retrieval within a unified Streamlit-based interface. Unlike cloud-dependent healthcare AI systems, the platform supports fully local deployment for most functionalities, enabling improved privacy, reduced latency, and accessibility on consumer hardware. The system utilizes MedGemma-4B for multimodal reasoning and LightRAG for contextual retrieval from structured electronic medical records generated using Synthea synthetic healthcare datasets. Experimental evaluation and qualitative analysis demonstrate the system's capability to generate clinically relevant responses for medical image interpretation and patient data querying. The project highlights the feasibility of deploying lightweight multimodal medical assistants in educational and research settings while addressing challenges related to hallucination, safety, and clinical reliability.**

***Index Terms*—Medical AI, MedGemma, Multimodal Models, Retrieval-Augmented Generation, Clinical Decision Support, Healthcare Informatics**

I. Introduction

Artificial intelligence has become increasingly important in modern healthcare systems due to its ability to assist clinicians in diagnosis, medical image interpretation, and patient data analysis. Recent progress in large language models (LLMs) and vision-language models (VLMs) has enabled the development of intelligent medical assistants capable of understanding both textual and visual medical data.

Despite these advances, many healthcare AI systems remain dependent on cloud infrastructure and proprietary APIs. Such dependence introduces limitations regarding privacy, latency, deployment costs, and accessibility in low-resource environments. In addition, many general-purpose language models lack domain-specific medical reasoning capabilities and often produce hallucinated or clinically unreliable outputs.

To address these challenges, this paper presents a lightweight multimodal medical assistant powered by MedGemma-4B and Retrieval-Augmented Generation (RAG). The proposed system combines local multimodal inference, medical image analysis, clinical question answering, and patient data retrieval within a unified platform. The architecture supports fully local execution for most functionalities using LM Studio and lightweight open-source models, enabling improved privacy and accessibility on consumer-grade hardware.

The primary contributions of this work are as follows:

- Design and implementation of a lightweight multimodal medical assistant using MedGemma-4B.
- Integration of Retrieval-Augmented Generation (RAG) for contextual clinical data retrieval.
- Support for local deployment using consumer hardware and LM Studio.
- Development of a unified Streamlit-based clinical support interface.
- Qualitative evaluation of multimodal medical reasoning and retrieval capabilities.

The implementation of the proposed system is publicly available as an open-source project at: https://github.com/srssina/MedGemma-Powered-Medical-Assistant .

II. Related Work

A. Medical Vision-Language Models

Recent advancements in multimodal foundation models have significantly improved the ability of artificial intelligence systems to process and reason over both medical images and clinical text. Traditional medical AI systems were primarily designed for narrow tasks such as image classification or disease detection, often requiring task-specific architectures and extensive supervised training. However, the emergence of large vision-language models (VLMs) has enabled more generalized medical reasoning capabilities across multiple modalities.

MedGemma represents one of the most recent developments in open medical vision-language models specifically designed for healthcare applications. The MedGemma technical report demonstrated strong performance across several multimodal medical benchmarks, including medical visual question answering, chest X-ray interpretation, and electronic health record retrieval tasks [1]. The model achieved measurable improvements over baseline Gemma architectures, particularly in multimodal reasoning and contextual medical understanding.

Similarly, LLaVA-Med introduced a biomedical adaptation of large language-and-vision assistants capable of processing clinical visual inputs and textual instructions [7]. The framework demonstrated that multimodal instruction tuning can significantly improve biomedical reasoning performance while reducing the need for extensive domain-specific retraining.

Other multimodal healthcare systems have focused on radiology and clinical image interpretation. CheXagent proposed a unified multimodal healthcare agent capable of performing chest X-ray analysis and radiology report generation using instruction-tuned medical vision-language architectures. Such systems highlight the growing importance of multimodal reasoning in modern clinical AI applications.

Compared to earlier specialized diagnostic systems, modern medical VLMs provide more flexible and generalized healthcare assistance capabilities. These models support conversational interaction, contextual reasoning, and integrated multimodal analysis, making them suitable for clinical support systems and educational healthcare applications.

B. Large Language Models in Healthcare

Large language models have become increasingly important in healthcare due to their ability to encode biomedical knowledge and generate context-aware clinical responses. Early transformer-based biomedical language models focused primarily on textual understanding tasks such as clinical note classification and biomedical entity recognition.

ClinicalBERT extended the BERT architecture for healthcare applications by training on clinical notes and hospital records [5]. The model demonstrated improvements in clinical prediction tasks and highlighted the importance of domain-specific language modeling for healthcare systems.

BioGPT further advanced biomedical natural language generation by introducing a transformer model trained specifically on biomedical literature [4]. The model showed strong performance in biomedical text generation, question answering, and medical summarization tasks.

PMC-LLaMA later introduced open-source large language models trained using biomedical and scientific literature from PubMed Central [6]. The project emphasized the importance of accessible open-source medical LLMs for healthcare research and educational applications.

One of the most influential developments in healthcare language models was Med-PaLM, which demonstrated that large language models could achieve strong performance on medical licensing examination-style benchmarks [3]. Med-PaLM highlighted the potential of foundation models for healthcare reasoning while also emphasizing concerns related to hallucination, reliability, and clinical safety.

Despite these advances, general-purpose medical LLMs often lack integrated multimodal capabilities and contextual patient retrieval systems. In addition, many healthcare AI platforms remain dependent on cloud-based APIs, limiting accessibility and introducing privacy concerns. The proposed system addresses these limitations by combining multimodal medical reasoning with lightweight local deployment and retrieval-augmented patient data access.

C. Retrieval-Augmented Generation in Healthcare

Retrieval-Augmented Generation (RAG) has emerged as an effective strategy for improving factual grounding in large language models. Traditional language models generate responses solely from parametric memory, which can lead to hallucinated or outdated information. RAG systems mitigate this issue by retrieving relevant contextual documents and incorporating them into the generation process.

Lewis et al. introduced one of the foundational RAG architectures for knowledge-intensive natural language processing tasks [2]. Their approach demonstrated that combining retrieval mechanisms with generative language models significantly improved factual accuracy and contextual consistency.

In healthcare applications, RAG systems have shown particular promise for electronic medical record analysis, clinical question answering, and medical knowledge retrieval. By incorporating patient-specific information during inference, RAG pipelines enable more personalized and context-aware response generation.

Several healthcare AI systems have adopted retrieval-augmented pipelines to reduce hallucination and improve clinical reliability. Recent studies have demonstrated that retrieval-based architectures can improve medical question answering performance while reducing factual inconsistencies in generated responses [9].

The proposed system integrates LightRAG as the retrieval framework for contextual patient data access. Synthetic electronic medical records generated using Synthea are embedded and indexed for efficient retrieval. Retrieved patient-specific context is injected into the MedGemma prompt to improve response grounding and reduce generic outputs.

D. Synthetic Healthcare Datasets

Access to real patient electronic medical records is heavily restricted due to privacy regulations, ethical concerns, and institutional approval requirements. As a result, synthetic healthcare datasets have become increasingly important for healthcare AI research and system evaluation.

Synthea is one of the most widely used synthetic healthcare data generation frameworks [10]. The platform generates realistic synthetic patient records, including demographics, diagnoses, medications, procedures, and longitudinal medical histories. Synthea has been extensively used in healthcare informatics research due to its ability to provide realistic yet privacy-preserving EMR datasets.

The use of synthetic datasets enables researchers to evaluate healthcare AI systems without exposing sensitive patient information. In the proposed system, Synthea-generated records were used to simulate electronic medical record retrieval workflows within the RAG subsystem.

E. Local Deployment and Privacy-Preserving AI

Recent research has increasingly emphasized the importance of privacy-preserving AI systems in healthcare environments. Cloud-dependent AI systems may expose sensitive medical data to external services and introduce latency and infrastructure limitations.

The emergence of lightweight open-source models and local inference platforms such as LM Studio has enabled researchers and developers to deploy advanced AI systems on consumer-grade hardware [11]. Local deployment provides several advantages for healthcare applications, including improved privacy, offline accessibility, and reduced operational costs.

The proposed architecture adopts a local-first deployment strategy in which MedGemma-4B inference is executed locally through LM Studio using an OpenAI-compatible API interface. This design improves deployment flexibility while reducing dependence on external cloud services.

Although certain large-scale retrieval operations still require external APIs due to hardware limitations, the majority of system functionalities operate locally. This architecture demonstrates the feasibility of deploying multimodal healthcare AI systems in low-resource and privacy-sensitive environments.

III. System Architecture

The proposed medical assistant consists of three primary components:

1) MedGemma-4B multimodal inference engine
2) Retrieval-Augmented Generation subsystem
3) Streamlit-based user interface

Figure 1 illustrates the overall system architecture.

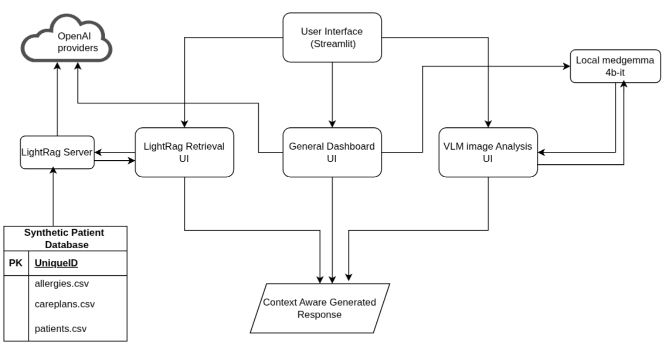

Fig. 1. Overall architecture of the MedGemma-powered medical assistant.

A. Medical Image Analysis Pipeline

The image analysis subsystem enables users to upload medical images such as radiology scans, dermatology images, pathology slides, CT scans, and X-rays. Uploaded images are processed by MedGemma-4B through LM Studio using an OpenAI-compatible inference API.

The model generates clinically relevant interpretations and descriptive reports based on multimodal reasoning capabilities.

B. Clinical Question Answering

The assistant supports general medical question answering through a conversational dashboard interface. User queries are processed using MedGemma-4B to generate context-aware healthcare responses.

The dashboard enables clinicians, nurses, and students to interactively access medical information using natural language.

C. Retrieval-Augmented Generation Pipeline

The RAG subsystem uses LightRAG to retrieve contextual patient information from structured datasets. Synthetic EMR data generated using Synthea is embedded using the `text-embedding-nomic-embed-text-v1.5` model.

The retrieval pipeline consists of:

1) Document ingestion
2) Embedding generation
3) Vector storage
4) Context retrieval
5) Response augmentation

Retrieved contextual data is injected into the language model prompt to improve factual grounding and reduce hallucination.

IV. Implementation

The system was implemented using Python 3.12 and several open-source frameworks.

A. Software Stack

Table I summarizes the primary technologies used in the implementation.

TABLE I
Software Stack

Component	Technology
Frontend UI	Streamlit
Vision-Language Model	MedGemma-4B
RAG Framework	LightRAG
Embedding Model	Nomic Embed v1.5
Inference Platform	LM Studio
Programming Language	Python 3.12

B. Local Deployment

Most functionalities are executed locally using LM Studio and open-source models. This architecture enables:

- Reduced dependency on cloud services
- Lower inference latency
- Improved data privacy
- Accessibility on consumer-grade hardware

However, OpenAI APIs are currently required for certain RAG ingestion operations due to my hardware limitations.

V. Experimental Evaluation

A. Evaluation Objectives

The experimental evaluation was conducted to assess the feasibility and effectiveness of the proposed multimodal clinical support system. The evaluation focused on three primary objectives:

- Assessing the multimodal reasoning capability of the underlying MedGemma-4B model.

- Evaluating the effectiveness of the Retrieval-Augmented Generation (RAG) subsystem for contextual patient data retrieval.
- Validating the practicality of lightweight local deployment using consumer-grade hardware.

The experiments were performed using synthetic electronic medical record (EMR) datasets generated with Synthea and publicly available medical imaging samples.

B. Reference Benchmark Performance

The MedGemma technical report demonstrated strong performance across several multimodal healthcare benchmarks [1]. These benchmark results provide contextual evidence regarding the capabilities of the underlying foundation model used in this work.

The benchmark results reported in this section originate from the official MedGemma technical report and are included to contextualize the capabilities of the underlying foundation model rather than to claim reproduction of the original experiments.

MedGemma achieved improvements across several medical tasks, including medical visual question answering, chest X-ray classification, agentic clinical evaluation, and electronic health record retrieval. Reported gains included 2.6–10% improvements on multimodal medical question answering tasks and 15.5–18.1% improvements on chest X-ray classification benchmarks compared to baseline Gemma models [1].

TABLE II
SELECTED MEDGEMMA BENCHMARK RESULTS FROM OFFICIAL TECHNICAL REPORT

Task	Dataset	Reported Improvement
Medical Visual QA	MedQA	2.6–10%
Chest X-ray Classification	ChestX-ray14	15.5–18.1%
Clinical Agent Evaluation	AgentClinic	10.8%
EHR Retrieval	Internal Benchmark	50% error reduction

These benchmark results support the suitability of MedGemma as the multimodal reasoning backbone for the proposed clinical support system.

C. System-Level Qualitative Evaluation

In addition to the reference benchmark analysis, qualitative experiments were conducted to evaluate the integrated workflow of the proposed system. The evaluation focused on image interpretation, patient record retrieval, and conversational medical question answering.

The experiments utilized:

- Synthetic patient records generated using Synthea.
- Publicly available medical imaging datasets.
- Local inference using LM Studio and MedGemma-4B.

D. Medical Image Analysis

The medical image analysis subsystem was evaluated using multiple categories of medical images, including chest X-rays, dermatology images, CT scans, and pathology samples.

The model demonstrated the ability to generate clinically relevant descriptions and observations for several imaging modalities. The generated outputs generally aligned with visible image characteristics and provided medically coherent explanations.

Table III summarizes the qualitative image analysis results.

TABLE III
QUALITATIVE MEDICAL IMAGE ANALYSIS RESULTS

Image Type	Task	Output Quality
Chest X-ray	Abnormality Description	Good
Dermatology	Lesion Interpretation	Good
CT Scan	Structural Observation	Moderate
Pathology	Tissue Description	Good

The results indicate that MedGemma-4B can provide useful preliminary interpretations for multimodal healthcare applications in educational and research-oriented settings.

E. Retrieval-Augmented Generation Evaluation

The Retrieval-Augmented Generation subsystem was evaluated using synthetic patient records and clinical question-answering tasks.

The LightRAG framework successfully retrieved relevant contextual patient information, including demographic data, diagnosis history, medications, and clinical observations. Retrieved information was incorporated into the language model prompt to improve contextual grounding.

F. Local Deployment Evaluation

The proposed architecture was designed to support lightweight local deployment using consumer-grade hardware through LM Studio.

Most functionalities, including multimodal image analysis and conversational medical assistance, were successfully executed locally without requiring cloud-based inference APIs. This deployment strategy provides several practical advantages:

- Improved data privacy
- Reduced latency
- Lower operational costs
- Offline accessibility

However, certain RAG ingestion operations required external API support due to hardware limitations during embedding generation and large-scale indexing.

VI. DISCUSSION

The proposed system demonstrates the feasibility of deploying lightweight multimodal medical assistants using open-source foundation models and local inference infrastructure.

A. Strengths

Key strengths of the system include:

- Lightweight deployment architecture
- Multimodal medical reasoning
- Integrated retrieval-based contextual grounding

- Privacy-friendly local execution
- Open-source reproducibility

B. Limitations

Several limitations remain:

- Lack of clinical validation
- Potential hallucination in generated responses
- Dependence on synthetic datasets
- Limited quantitative benchmarking

Future work may include fine-tuning using specialized medical datasets, integrating larger models, and conducting expert clinical evaluations.

VII. Ethical Considerations

This project is intended solely for educational and research purposes and is not approved for clinical use. The system should not replace licensed healthcare professionals or medical diagnosis procedures.

All patient data used in experiments was synthetically generated using Synthea, and no real patient electronic medical records were utilized.

The system may generate inaccurate or hallucinated medical information, and outputs should always be reviewed by qualified medical personnel.

VIII. Conclusion

This paper presented a lightweight multimodal medical assistant powered by MedGemma-4B and Retrieval-Augmented Generation. The proposed system integrates medical image analysis, conversational clinical support, and contextual patient data retrieval within a unified platform.

The architecture demonstrates the feasibility of deploying privacy-friendly medical AI systems locally on consumer hardware while leveraging multimodal foundation models for healthcare assistance. Experimental evaluation showed promising performance for educational and research-oriented clinical support tasks.

Future work will focus on quantitative benchmarking, clinical validation, larger multimodal models, and enhanced safety mechanisms for healthcare deployment.

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